20/1/26 Recitation: 2

A mobile service provider wants to classify SMS messages as Spam or Not Spam using a single-layer neural network. With 5 hidden neurons in the layer.

x_1	x_2	x_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
-2	1	1	0

	w_1	w_2	b
h_1	0.1	0.2	1
h_2	-0.2	0.3	1
h_3	0.4	-0.1	1
h_4	0.3	0.2	1
h_5	0.2	-0.4	1

(i) $x_1 = 4$, $x_2 = 2$, $b = 1$

$w_1 = 0$, $w_2 = 0$

$z_1 = w_1 x_1 + w_2 x_2 + b$

$= \cancel{0.4 + 0.2 + 1 = 1.8}$

$\cancel{z = 1, \hat{y} = 1 = y}$

$= (0.1)4 + (0.2)2 + 1$

$0.4 + 0.4 + 1 = \boxed{1.8}$

$Z_2 = w_1 x_1 + w_2 x_2 + b$

$= 4(-0.2) + (0.3)(2) + 1$

$-0.8 + 0.6 + 1 = \boxed{0.8}$

$Z_3 = 0.4(4) + (-0.1)2 + 1$

$= 1.6 - 0.2 + 1$

$= \boxed{2.4}$

$Z_4 = 0.3(4) + 0.2(2) + 1$

$= 1.2 + 0.4 + 1 = \boxed{2.6}$

$Z_5 = 0.2(4) + (-0.4)2 + 1$

$0.8 + (-0.8) + 1 = \boxed{1}$

$a(Z_1) = \frac{1}{1 + e^{-Z_1}} = 0.858$

$0.86 = h_1$

$a(Z_2) = \frac{1}{1 + e^{-0.8}} = 0.689$

$= 0.69 = h_2$

$a(Z_3) = \frac{1}{1 + e^{-2.4}} = 0.916$

$0.92 = h_3$

$a(Z_4) = \frac{1}{1 + e^{-2.6}} = 0.93 = h_4$

$a(Z_5) = \frac{1}{1 + e^{-1}} = 0.73 = h_5$

$v_1 = 0.1$, $v_2 = (-0.2)$, $v = 0.3$

$v_4 = 0.4$, $v_5 = (-0.3)$

$$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_4 v_5$$

$$= 0.086 + (-0.138) + 0.276$$

$$+ 0.372 + (-0.219)$$

$$= 0.377 \approx 0.38$$

$$a(z_{out}) = 0.59$$

$$0.6$$

$$\frac{1}{1+e^{z_{out}}}$$

$$\hat{y} = \begin{cases} 0 & a(z_{out}) < 0.5 \\ 1 & a(z_{out}) \geq 0.5 \end{cases}$$

$$\hat{y} = 1 \approx y$$